

Kuan-Ting (Larry) Chen

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Education

National Yang Ming Chiao Tung University, [Advanced Database System Laboratory](#)

Hsinchu, Taiwan

M.S. in Computer Science and Engineering (Advisor: Prof. Wen-Chih Peng)

2024/09 – Present

- Research: Sports Analytics, Large Language Models, and Reinforcement Learning
- GPA: 4.3/4.3; 2 Academic Excellence Awards; 2 publications; Member, The **Phi Tau Phi** Scholastic Honor Society

National Cheng Kung University, [Intelligent Knowledge Management Laboratory](#)

Tainan, Taiwan

B.S. in Computer Science and Information Engineering (Advisor: Prof. Hung-Yu Kao)

2020/09 – 2024/06

- Research: Natural Language Processing, Hateful Meme Detection, and Fact Checking
- GPA: 4.21/4.3; 1 Academic Excellence Award; 3 extracurricular awards

Experience

Student Project Coordinator, [Precision Sports Science Research Project](#)

2025/05 – 2026/09

- Coordinated integration across 5 subprojects, preparing 3 consolidated reports and facilitating 2 project demos.
- Developed **BadmintonTwin**; co-developed **CoachAI+** and **BadmintonReels**, reducing highlight production from ~1 day to ~5 minutes; contributed to **BadmintonGPT** system design, integrating badminton data, web search, and 4 MCP tools.

Projects

BadmintonTwin: An LLM-based Virtual Badminton Player

[Code](#)

- Developed a multi-agent LLM framework with hybrid RAG, integrating physical-state retrieval and semantic contextual memory to enable controllable, player-specific badminton rally simulation.
- Achieved the best overall simulation fidelity among 7 baselines across landing, movement, and shot-type metrics on ShuttleSet Series; expert evaluation further confirmed strong tactical alignment and realism of generated rallies.

Skills: Large Language Models · Retrieval-Augmented Generation · AI Agents · Imitation Learning

CoachAI+ Badminton Environment

[Code](#)

- Developed PDQN, HybridPPO, and HybridSAC baselines for hybrid action-space RL in badminton strategy optimization.
- Designed simulation environments and evaluation metrics by integrating coach knowledge into quantifiable objectives.

Skills: Reinforcement Learning · Hybrid Action Space · Simulator · PyTorch · Data Analysis

Medical Conversation Analysis System

[Code](#)

- Built a dual-server pipeline leveraging speaker diarization, Azure ASR, and medical NER to automatically generate doctor-patient transcripts and extract key clinical terms for physician review.

Skills: Python · Flask · SQL · Speaker Diarization · Azure Speech Services · Named Entity Recognition

Publications

Kuan-Ting Chen, Kuang-Da Wang, and Wen-Chih Peng. “BadmintonTwin: Tactical Memory-Augmented LLM Agents for Instruction-Conditioned Badminton Rally Simulation.” **Under review at ACL Rolling Review.**

Tsan-Tsung Yang, I-Wei Chen, **Kuan-Ting Chen**, Shang-Hsuan Chiang, and Wen-Chih Peng. “Robust Detection and Source Identification of AI-Generated Images Using CNN and CLIP-Based Models.” **AAAI Defactify 2025.** [[Paper](#)] [[Code](#)]

Awards

College Student Research Creativity Award

[Code](#) 2024/07

- Awarded to only ~5% of participants. Converted multimodal inputs and HurtLex knowledge into text, then finetuned HateBERT with Masked Language Modeling (MLM) objective function.

AI Cup 2023 Factual Text Retrieval and Verification Competition – Honorable Mention

[Code](#) 2024/03

- Achieved 13th place out of 301 participants. Built a 3-stage fact verification pipeline with multi-source retrieval and BERT-based evidence selection/claim verification, improving ACC by 22.7 percentage points over the official baseline.

Meichu Hackathon Hacker Group (TSMC) – Second Place

[Code](#) 2022/10

- Co-developed the prototype within a 48-hour hackathon. Finetuned YOLOv7 with transfer learning and strong data augmentation, then deployed real-time inference on laptops and Raspberry Pi with REST API integration.

Technical Skills

Proficient: Python, PyTorch, Agentic AI, NLP, Reinforcement Learning, Machine Learning, vLLM

Familiar: C++, Linux, Computer Vision, Parallel Programming, Graph Neural Networks, CI/CD

Languages: Mandarin (Native), English (Upper-intermediate; TOEIC: 800)